

Brenna Ren

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EDUCATION

Yale University

Bachelor of Science, Computer Science and Mathematics

New Haven, CT

May 2030

The Harker School

GPA: 4.00/4.00 (UW), 4.65/4.80 (W) • SAT: 1570 (770 RW, 800 M)

San Jose, CA

May 2026

- *Relevant Coursework:* Data Structures, Neural Networks, Computer Architecture, Robot Kinematics Software, Compilers & Interpreters, Linear Algebra, Discrete Math, Multivariable Calculus, Differential Equations
- *Activities:* Programming Club President, AP CS & Compilers TA, Technology Student Association President

EXPERIENCE

Data Research Intern

College Kickstart

June 2026 – Present

Remote

- Migrated assured admissions checker to a **database-driven eligibility engine**, enabling updated policy changes for counselors and students at hundreds of U.S. high schools
- Improved list reliability for counselors and students by verifying **20K+ data points** across 750 colleges; authored a top-40 robotics schools blog read by thousands of readers

Executive President

FRC Team 1072: Harker Robotics

August 2022 – May 2026

San Jose, CA

- Reduced robot cycle times by **~20%** by implementing driver-assist **auto-alignment** using real-time pose estimation from a Kalman filter fusing odometry, gyro, and vision data
- Achieved **2x World Championship Qualification** and 2x Offseason Champions by directing end-to-end robot development across a **102-member team** with a **\$50K budget**
- Maintained a **20K LOC Java control codebase** with 150+ personal Git commits
- Grew membership **15%** by leading 100+ annual meetings; earned FIRST Dean's List Semi-Finalist for leadership

Garcia Summer Research Scholar

Stony Brook University

June 2024 – March 2026

Stony Brook, NY

- Published **first-author manuscript in MRS Communications** on an ML model predicting thin-film behavior from experimental data; DOI: 10.1557/s43579-026-00956-5
- Achieved **$R^2 = 0.9919$** on held-out samples by building an end-to-end **Python** pipeline to process, clean, and model **300+** experimental measurements
- Communicated findings in poster and oral presentations at MRS, ACS, and APS international conferences

PROJECTS

Deep Learning for Medical Signal Analysis | *Python, Tensorflow/Keras, Git*

August 2024 – June 2025

- Improved RBD diagnostic accuracy from **80% to 97%** by designing an unsupervised deep learning model to detect the condition from ECG data
- Published as first author in **Springer SCIS (top 15 papers)**; presented at the 23rd IEEE/ACIS SERA conference; DOI: 10.1007/978-3-032-06658-9_7

Pascal-Subset Compiler & Interpreter | *Java, MIPS Assembly*

January 2024 – May 2024

- Built a full compiler and AST-walking interpreter in **Java** for a Pascal subset, implementing a hand-coded scanner, recursive descent parser, semantic analysis, and **MIPS assembly** code generation
- Engineered stack frame management to support **recursive procedures** and nested variable scoping in generated machine code, verified on a MIPS simulator

AWARDS AND HONORS

USACO Gold Division • AIME Qualifier • NCWIT Aspirations in Computing (2x National Honorable Mention) • ACSL Senior Division Bronze Medal (3x Finals Qualifier)

TECHNICAL SKILLS

Languages: Java, C++, Python, MIPS Assembly

Tools & Frameworks: Git, Linux, TensorFlow, Keras, Qiskit, Mathematica